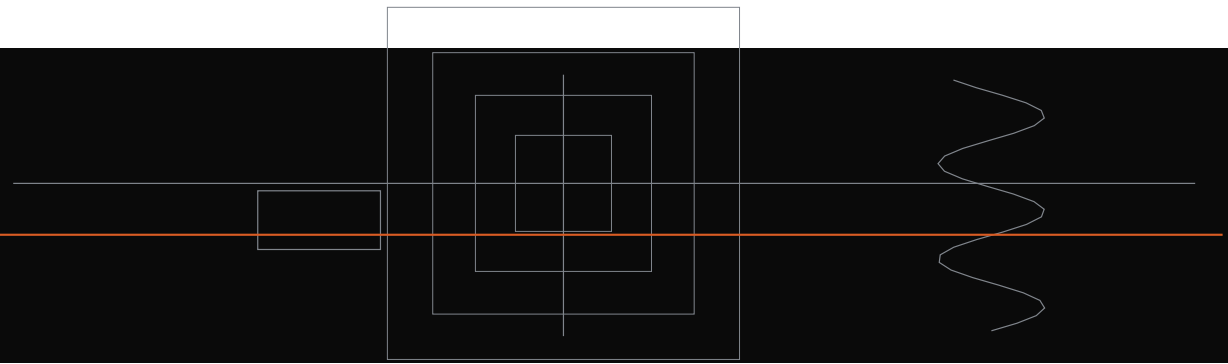


BILL OF MATERIALS

Quadruped I

Rev. 2026-04-30 67 items



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BOM

Product/Project: Quadrupe I
Revision date: 2026-04-30

<https://www.printables.com/model/1443108-quadrupe-i>

LEG LINKAGE MECHANISM

11 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Servo Motors	x12	AliExpress Link	Amazon Link	180 degrees, 35kg (Approx 49.5x20,2x40mm)
>>	Raspberry Pi 4 Model B	1	AliExpress Link	Amazon Link	8gb Ram
>>	LiPo Battery	1	AliExpress Link	Amazon Link	4000mAh 4S 45~90C Lipo Pack w/XT-90 (You can change to the XT-90 connect later)
>>	IMU	1	AliExpress Link	Amazon Link	Integrated Measuring Unit
>>	Raspberry Pi Breakout Board	1	AliExpress Link	Amazon Link	Made by Dfrobot
>>	Fan	x2	AliExpress Link	Amazon Link	-
>>	Servo Driver PCA9685	1	AliExpress Link	Amazon Link	To control servo motors, via I2C interface
>>	Arduino Nano	1	AliExpress Link	Amazon Link	For IMU, Battery sensing, LCD screen
>>	DC to DC (Buck) converter 20A	1	AliExpress Link	Amazon Link	Adjustable, set to ~7V in Dingo
>>	Power Distribution PCB	1	Order PCB HERE		Make sure to chose PCB + Assembly
>>	Orange Acrylic	1	AliExpress Link	Amazon Link	Adding STEP + DXF in case you cant print this.

BOM	Product/Project: Quadrupe I Revision date: 2026-04-30	https://www.printables.com/model/1443108-quadrupe-i
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BODY CONSTRUCTION 4 items					
	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	LCD Screen	1	AliExpress Link	Amazon Link	<i>1.47 inch</i>
>>	DC to DC (buck) converter 3A	1	AliExpress Link	Amazon Link	<i>For fans</i>
>>	Emergency Stop Switch	1	AliExpress Link	Amazon Link	<i>3a at 12-220V rated</i>
>>	32GB MicroSD card	1	AliExpress Link	Amazon Link	<i>32GB is maximum Pi supports</i>

BOM	Product/Project: Quadrupe I Revision date: 2026-04-30	https://www.printables.com/model/1443108-quadrupe-i
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COMPUTING 7 items					
	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Bearing	x4	AliExpress Link	Amazon Link	6705-2RS 25X32X4
>>	Thrust Bearings (legs)	1	AliExpress Link	Amazon Link	F4-10M 4x10x4mm
>>	Bearing	x20	AliExpress Link	Amazon Link	684ZZ Mini Deep Groove Ball Bearing 4x9x4mm
>>	Servo horns	x4	AliExpress Link	Amazon Link	RC Car Steering Servo Horn Arm 25T Teeth Long
>>	DC to DC (Buck) converter 4A	1	AliExpress Link	Amazon Link	For all components except servos and fans
>>	RasbPi Cooling System	1	AliExpress Link	Amazon Link	-
>>	Arduino Nano Breakout Board	1	AliExpress Link	Amazon Link	-

BOM	Product/Project: Quadrupe I	https://www.printables.com/model/1443108-quadrupe-i
	Revision date: 2026-04-30	

POWER SUPPLY AND MANAGEMENT 4 items					
	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Terminal Blocks	x8	AliExpress Link	Amazon Link	1935776
>>	Heatsinks	x4	AliExpress Link	Amazon Link	6mm, black. Cut in half for each leg motor
>>	GPIO Cable	1	AliExpress Link	Amazon Link	To connect Pi to its breakout board
>>	Servo Discs	x10	AliExpress Link	Amazon Link	with screws.

BOM

Product/Project: Quadrupe I
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<https://www.printables.com/model/1443108-quadrupe-i>

PCB PARTS

10 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Magnets	x8	AliExpress Link	Amazon Link	Used to hold fans in place
>>	Rod ends,Long	x4	AliExpress Link	Amazon Link	S 3mm L30mm M3
>>	Rod long	x4	AliExpress Link	Amazon Link	90mm long (cut if needed)
>>	Rod ends Short	x4	AliExpress Link	Amazon Link	N 3mm L20mm M3
>>	Rod short	x2	AliExpress Link	Amazon Link	16mm long (cut if needed)
>>	Logic Level Converter	1	AliExpress Link	Amazon Link	For adjusting voltage between the pi and the nano
>>	M3 threaded inserts	1	AliExpress Link	Amazon Link	Pack of 100, M3xD4.6xL4.0
>>	XT60H Connector	1	AliExpress Link	Amazon Link	To connect LiPo to the robot
>>	On/off Switch	1	AliExpress Link	Amazon Link	20A at 14VDC rated
>>	Spring	1	AliExpress Link	Amazon Link	For battery latch mechanism. 35mm long x 7mm OD

BOM

Product/Project: Quadrupted I
Revision date: 2026-04-30

<https://www.printables.com/model/1443108-quadrupted-i>

ELECTRICAL

9 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Male Connector (for peripherals)	x12	AliExpress Link	Amazon Link	B2B-PH-K-S(LF)(SN)
>>	Digital Voltmeter	1	AliExpress Link	Amazon Link	DC 2.5-30V, used to measure battery voltage
>>	Socket for Connectors	x25	AliExpress Link	Amazon Link	SPH-002T-P0.5S
>>	Threaded rods, long	x5	AliExpress Link	Amazon Link	M3 x 110mm
>>	Female Connector (for peripherals)	x12	AliExpress Link	Amazon Link	PHR-2
>>	Capacitor	1	AliExpress Link	Amazon Link	2200uF 16VDC Electrolytic RB Capacitor ,For servo board
>>	Threaded rods,short	x5	AliExpress Link	Amazon Link	M3 x 25mm
>>	Reverse Polarity Protection Diode	1	AliExpress Link	Amazon Link	B160-13-F
>>	47k Resistor	1	AliExpress Link	Amazon Link	CRGCQ1206F47K

BOM

Product/Project: Quadruped I
Revision date: 2026-04-30

<https://www.printables.com/model/1443108-quadruped-i>

MISC

9 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	10k resistor	1	AliExpress Link	Amazon Link	CRGCQ0805F10K
>>	PS4 Controller	1	AliExpress Link	Amazon Link	-
>>	Power Delivery Wire	1	AliExpress Link	Amazon Link	Minimum 12AWG, for wires expected to carry current above 2A and/or voltage above 5V
>>	Connection Wire	1	AliExpress Link	Amazon Link	22-30AWG, recommend solid core, used for all other wires
>>	Ferrules	1	AliExpress Link	Amazon Link	Optional, if used need approx 20 across robot
>>	Heat Shrink	1	AliExpress Link	Amazon Link	Only small amount required, for covering exposed solid-core wires
>>	100k resistors	1	AliExpress Link	Amazon Link	For dividing the voltage at the buck converter so that it is readable by the nano
>>	M3 Nuts	1	AliExpress Link	Amazon Link	-
>>	M3 Bolts	1	AliExpress Link	Amazon Link	-

BOM	Product/Project: Quadruped I Revision date: 2026-04-30	https://www.printables.com/model/1443108-quadruped-i
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FILAMENTS 4 items					
	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Black ABS/ASA/PETG&PC Filament	1	AliExpress Link	Amazon Link	Approx 500g required
>>	Yellow ABS/ASA/PETG&PC Filament	1	AliExpress Link	Amazon Link	Approx 500g required
>>	Black PLA Filament	1	AliExpress Link	Amazon Link	Approx 750g required (includes 500g for the stand)
>>	White TPU Filament	1	AliExpress Link	Amazon Link	Approx 50g required, for feet

BOM	Product/Project: Quadruped I Revision date: 2026-04-30	https://www.printables.com/model/1443108-quadruped-i
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TOOLS 9 items					
	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Crimp Tool	1	AliExpress Link	Amazon Link	-
>>	Hex Key set	1	AliExpress Link	Amazon Link	-
>>	Soldering Iron	1	AliExpress Link	Amazon Link	-
>>	Wire cutter	1	AliExpress Link	Amazon Link	-
>>	Wire stripper	1	AliExpress Link	Amazon Link	-
>>	Precision Screwdriver Set	1	AliExpress Link	Amazon Link	-
>>	Heat insert set	1	AliExpress Link	Amazon Link	-
>>	Heat insert tip - set	1	AliExpress Link	Amazon Link	-
>>	Soldering Iron TS101	1	AliExpress Link	Amazon Link	-